Music Intelligence with Deep Learning

Noah Wiley | CSCI 198 Senior Project | California State University, Fresno
Supervisor: David Ruby

Problem Statement

Music classification identifies genre, emotion, or instrumentation from audio and remains a core open problem in Music Information Retrieval (MIR). While CNNs, LSTMs, and Transformers have individually shown promise, there is limited comparative work in an undergraduate setting that unifies these architectures under a single framework. This project designs, trains, and evaluates CNN, BiLSTM, and Transformer models on three publicly available music datasets — GTZAN, FMA Small, and MagnaTagATune — and builds an interactive demo application for real-time audio classification.

Datasets

GTZAN

1,000 clips, 10 genres, 30s each, single-label classification

FMA Small

~8,000 clips, 8 genres, cross-dataset benchmarking

MagnaTagATune

~25,000 clips, 188 binary tags, evaluated with ROC-AUC

Preprocessing Pipeline

- Resample to 22,050 Hz
- Standardize to 30s (pad/trim)
- 128-bin log-mel spectrogram
- Normalize (zero mean, unit variance)

Related Work

- AMAI Lab: deep learning approach for emotion/genre
- CNNs capture spatial spectrogram patterns
- LSTMs model temporal dependencies
- Transformers analyze long-range structure

Model Architectures

CNN

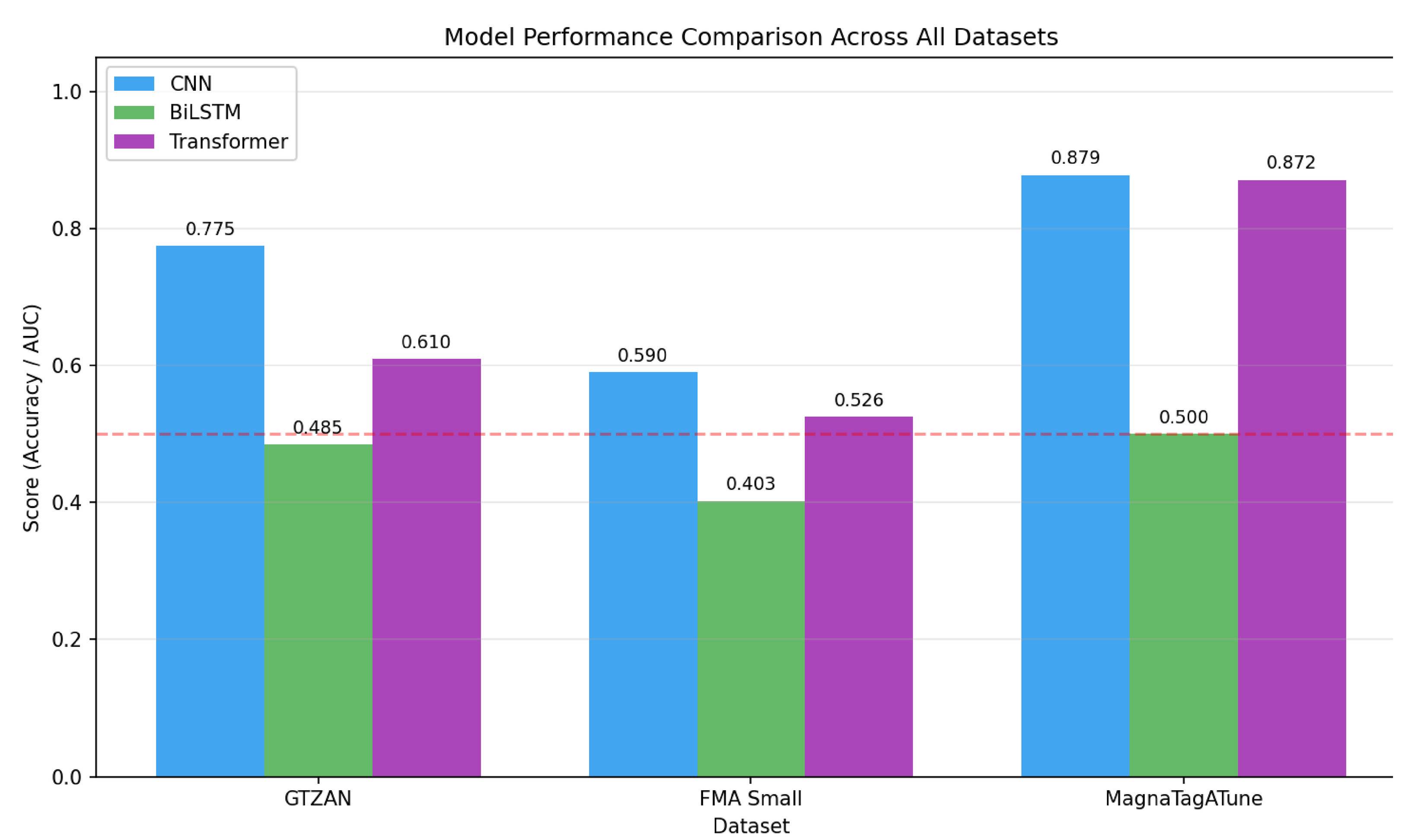
- 4-layer CNN with batch norm, ReLU, max pool, dropout
- Treats spectrogram as 2D image
- CrossEntropy / BCE loss

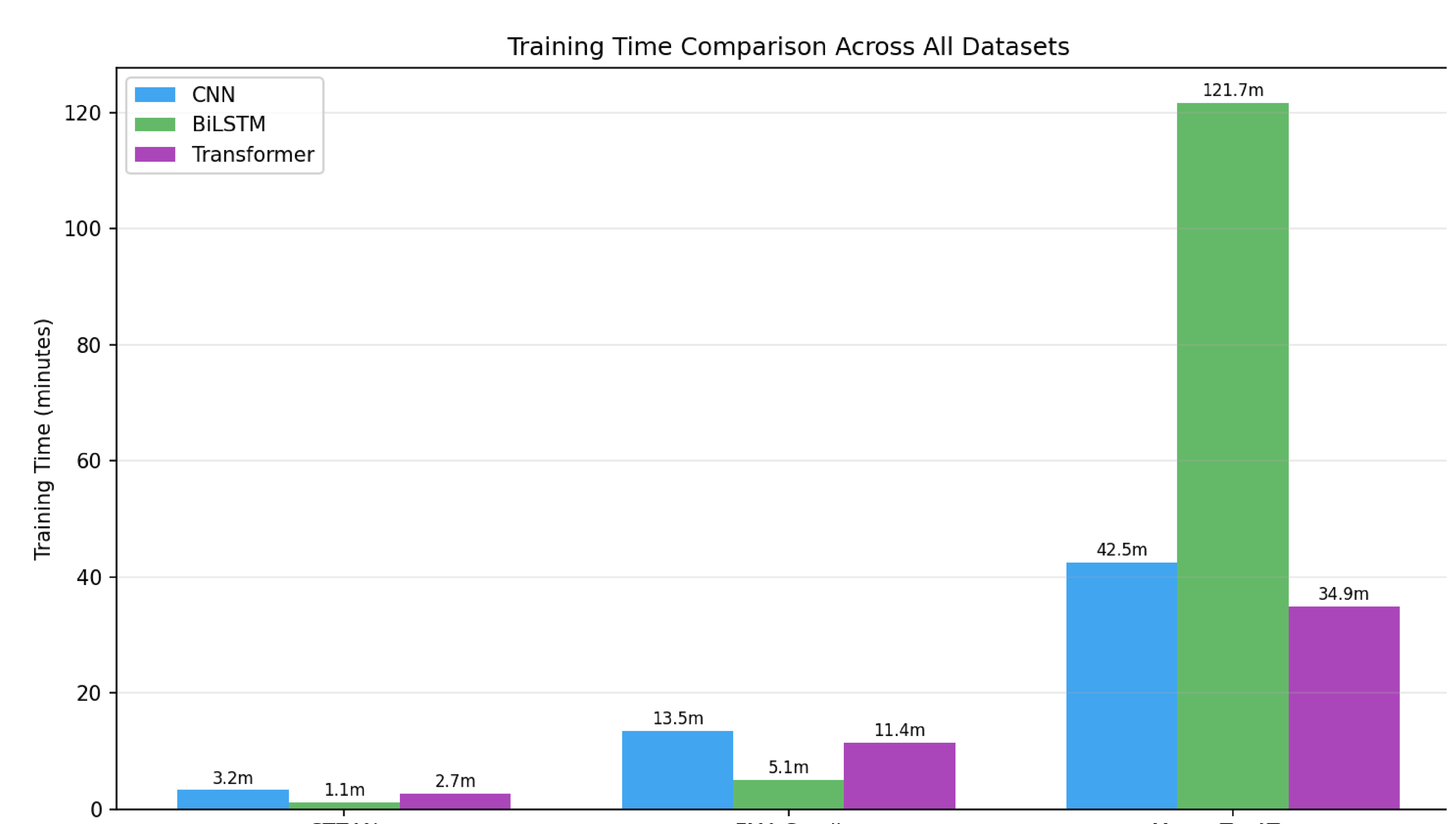
BiLSTM

- 2-layer bidirectional LSTM
- Processes time frames forward and backward
- Captures temporal dependencies

Transformer

- Multi-head self-attention with positional encodings
- 4 attention heads, 2 encoder layers
- Captures long-range dependencies





Key Findings

- CNN outperformed all models on all 3 datasets
- Transformer consistently second; BiLSTM last
- BiLSTM failed on MagnaTagATune (AUC 0.50)
- Transformer predicted 4x more tags than CNN
- Rock and Pop hardest genres across all models

Demo App

- Built with Streamlit (Python)
- Upload mp3/wav for real-time predictions
- Displays spectrogram and confidence scores

Limitations

- No data augmentation
- Limited training epochs
- GTZAN small size may cause overfitting

Future Work

- Add SpecAugment and data augmentation
- Explore CNN + Transformer ensembles
- Fine-tune pre-trained models (HuBERT, Wav2Vec2)
- Deploy demo via Streamlit Cloud

Conclusion

- CNNs most effective for spectrogram-based classification
- Transformers excel at multi-label tagging
- BiLSTMs better for simpler sequential tasks
- All code and demo available on GitHub